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MILITARY SPECIFICATION

DETECTOR, ANTI-INTRUSION: DT-383/GSQ (AAU)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an add-on acoustic detection device to be used as an ancillary unit to the MINISID III.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-C-45662	- Calibration System Requirements.
MIL-H-46855	- Human Engineering Requirements for Military System, Equipment and Facilities.

FSC 6350

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13218E8610	- Detector Anti-Intrusion: DT-383/GSQ (AAU)

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Detector Anti-Intrusion DT-383/GSQ (hereinafter called AAU) shall be in accordance with Top Assembly TA13218E8610 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow

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sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more AAU's for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce AAU's that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more AAU's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Inherently fungus-resistant or fungus-proofed materials shall be used where applicable. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.5 Environmental requirements.

3.5.1 Operating. The AAU shall operate as specified herein without malfunction or damage (see 6.4.1) while being exposed and after exposure to the following conditions:

(a) Temperature range: Plus 32° F (0° C) to plus 145° F (63° C).
(b) Altitude: 0 to 14,000 feet.
(c) Humidity: Up to 100 percent.
(d) Salt fog (see 4.6.2.11).
(e) Immersion: When tested as specified in 4.6.2.15, the AAU shall not be damaged while operating under 36 inches of water. The AAU shall show no evidence of leaks or electrical or mechanical malfunction and shall be capable of operation as specified herein after removal from the water. The microphone shall not be submersed.

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3.5.2 Non-operating. The AAU shall operate as specified herein without malfunction or damage (see 6.4.1) after exposure to the conditions listed below. With exception of the temperature shock test, the AAU shall be in its individual carrying case during this exposure.

(a) Temperature range: Minus 65° F (minus 54° C) to plus 160° F (71° C).
(b) Altitude: 0 to 50,000 feet.
(c) Humidity (see 4.6.2.10).
(d) Temperature shock: 160° F / -65° F (see 4.6.2.6).
(e) Shock (see 4.6.2.9).
(f) Vibration (see 4.6.2.8).
(g) Fungus growth (see 4.6.2.12).
(h) Salt fog (see 4.6.2.11).

3.6 Performance. The AAU shall provide audio verification of an intrusion incident for a 15-second period following the seismic detection of the intrusion sensed by the MINISID III. The daylight sensor shall permit the unit to be inoperative only during the daylight hours. The sensing element shall be capable of discriminating between light emitted from sources such as battlefield flares, searchlights, etc., and natural daylight even when the daylight passes through foliage, clouds, smoke, and other filtering agents. The AAU shall also be suitable for use as an auxiliary battery for the MINISID III.

3.6.1 Modular or functional elements. The AAU shall consist of the following assemblies and subassemblies:

(a) Microphone assembly.
    (1) Microphone (see 3.6.1.1).
    (2) Daylight sensor (see 3.6.1.2).
    (3) Cable and plug assembly.
(b) Electronics assembly.
    (1) Amplifier and keyed A.G.C. (see 3.6.1.3).
    (2) Function control circuitry (see 3.6.1.4).
    (3) Logic and interface circuitry (see 3.6.1.5).
    (4) Electronics case assembly.

3.6.1.1 Microphone. Overall audio sensitivity shall be such that a sound pressure of 37 db above 0.0002 microbar will generate an AAU output signal to MINISID III of 300 millivolts rms, plus or minus 20 percent.

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3.6.1.2 Daylight sensor. The daylight sensor incorporated into the microphone housing shall inactivate the AAU during daylight hours. The movable light shutter shall provide for deactivation of the daylight sensor.

3.6.1.3 Amplifier and keyed AGC. The keyed AGC in the audio amplifier shall restrict the dynamic range of the output signal to MINISID III while preserving amplitude information. To minimize system distortion, the output signal shall be limited to not more than 1.0 volts rms. In order to preserve background noise level information and to allow for judgement of character and range of disturbances, the amplifier shall deliver an output of 300 millivolts rms to MINISID III. Background noise level shall be sampled for 300, plus or minus 50, milliseconds, during which the amplifier gain shall be automatically adjusted to the required level. Gain shall be locked at this level at the end of the sample interval and this gain setting shall be maintained within plus or minus 5 percent for the duration of the audio transmission (15 sec, plus or minus 10 percent, at ambient temperature and 15 sec, plus or minus 20 percent, at temperature extremes). The upper limit of the output amplitude (1.0V rms) shall be obtained by a minimum distortion limiter which acts only near the 1.0V limit. The amplifier response over the range resulting in output signals from 0.3 to 1.0V rms shall be linear with a total combined distortion not exceeding 5 percent. Frequency response of the overall system including the microphone shall be within plus or minus 7 db from 50 to 2000 Hz.

3.6.1.4 Function control circuitry. The interconnection diagram from the AAU/MINISID III system is shown in Figure 1. When the AAU connector is physically disconnected from MINISID III, automatic switch-over to seismic operation shall be accomplished in MINISID III. The movable light shutter is the only control on the AAU (see 3.6.1.2). Switch-over from daylight to nighttime operation shall be accomplished within the AAU.

3.6.1.5 Logic and interface circuitry. The MINISID III operates normally when no ancillary device is connected but shall be switched to "AND" operation when the AAU ancillary device is connected. When the AAU is connected, a positive going pulse (see 3.6.1.6) termed MINISID Detect Signal will be sent to the AAU when MINISID senses a valid seismic disturbance. Receipt of three of these activation pulses within 29, plus or minus 1.5, sec will initiate the 300 millisecond timer which shall determine the background sensing and AGC set interval. At the end of this interval, a 15 second, plus or minus 10 percent, timer shall be started. This interval shall determine the duration of audio transmission

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by MINISID III by sending the Audio Time signal shown in Figure 1 and specified in 3.6.1.6. Concurrent with the Audio Time signal, the amplifier output signal shall be sent to the MINISID III - modulator on the line labeled Audio Analog in Figure 1. Characteristics of this output signal are specified in 3.6.1.6. At the end of the 15-second audio transmission interval, the AAU shall completely deactivate its circuitry and shall be ready to accept another activation from MINISID III no later than 100 milliseconds after shutdown. When ambient light conditions in the vicinity of the daylight sensor in the microphone are typical of natural daylight from near sunrise to near sunset, the AAU shall not accept activation signals from MINISID III and this shall cause the AAU to cease transmission of even valid intrusion detections during daylight hours. It shall be possible to override this feature by the movable cover specified in 3.6.1.2.

3.6.1.6 Characteristics of interface signals. Characteristics of interface signals shall be as follows (see Figure 1):

(a) MINISID Detect Signal - A positive pulse of 5.6 to 6.3V amplitude, duration of 15, plus or minus 5, msec, and rise and fall time less than 0.3 msec (0.1V to 5.6V) on pin D.
(b) Audio Time - A positive pulse of 5.6 to 6.3V amplitude and duration of the intended audio transmission time (15 sec) on pin H.
(c) Audio Analog - An audio signal, frequency response to 50 Hz, plus or minus 10 percent, to 2000 Hz, plus or minus 10 percent, flat within plus or minus 7db; a sound pressure 37 db above a 0.0002-microbar reference level shall produce a 0.3V rms audio analog. The amplitude of the audio analog shall vary linearly with the input at the microphone and shall be a maximum of 1V rms on pin F.
(d) Encoder Callup Signal - A positive 5.5 to 6.3V dc pulse of 15, plus or minus 5, millisecond duration with a maximum rise and fall time of 0.3 milliseconds as measured into a 1 Kohm load on pin A.

3.6.1.7 Voltage operation range. The AAU shall operate as specified herein with a standby current drain $\leq$ 1.2 mA and an operating current $\leq$ 28 mA for battery voltages from 23 to 30 volts.

3.6.1.8 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the AAU shall conform to MIL-STD-461, Class I.

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3.6.1.9 Burn-in. Each AAU delivered to the Government shall be operated without failure for 48 consecutive hours with 100 activations per day.

3.6.2 Period and condition and storage. The AAU, excluding the battery, shall have a minimum shelf life of three years when stored at a temperature within the range of 63° F to 77° F.

3.7 Reliability. The AAU shall have a specified mean-time-between-failure (MTBF) of 6834 hours and an acceptable MTBF of 2278 hours, assuming 100 activations per day. When specified (see 6.2), reliability testing shall be conducted in accordance with 4.6.2.17.

3.8 Government-loaned property. The Government will loan the following for test purposes (see 6.5):

(a) MINISID III AN/GSQ-154.
(b) Batteries BA 1546/U.
(c) Receiver AN/USQ-46A.

3.9 Operation. Any changes made to the AAU shall include human engineering design characteristics. Changes shall be designed to conform to the applicable human engineering design criteria as specified in MIL-STD-1472 and MIL-H-46855.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. These requirements applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, freedom of parts from burrs, sharp edges, or any other damage or defects that could make the part unsuitable for the purpose intended.

3.10.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The AAU shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The AAU shall be subjected to the tests marked "X" in Column 1 of Table I. Failure of any test shall be cause for conducting the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production AAU's will be selected at random by the Government from AAU's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production detecting set shall not exclude the remaining AAU's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.4.1 Inspection failure. Failure of an initial production AAU to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production AAU and shall be cause for refusal by the Government to continue acceptance of production AAU's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on AAU's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all AAU's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. AAU samples selected in accordance with 4.5.1 shall be examined for the possible defects specified in 4.6.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each AAU shall be subjected to the tests marked "X" in Column 3 of Table I. Failure of any test shall be cause for rejection.

4.5.3.2 Sample. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table I. AQL shall be 1.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The AAU sample shall be examined as specified herein for the following possible defects:

Major

101. Metals not as specified.
102. Printed wiring assemblies and boards not as specified.
103. Aluminum castings not as specified.

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104. Microphone assembly not as specified.
105. Electronics assembly not as specified.
106. Construction not as specified.
107. Nonconformance to any requirement shown on the drawings.
108. Fungus and moisture resistance not as specified.
109. Protective treatment not as specified.

Minor

201. Treatment and painting not as specified.
202. Markings not as specified.
203. Workmanship not as specified.

TABLE I. Test schedule.

Pre-production	Initial Production and Inspection Comparison	Quality Conformance Individual	Quality Conformance Samples	Test	Requirement Paragraph	Test Paragraph
1	2	3	4	5	6	7
X	-	X	-	Burn-in.	3.6.1.9	4.6.2.2
X	X	X	-	Unit test.	3.6	4.6.2.3
X	X	-	-	High operating temp.	3.5.1(a)	4.6.2.4
X	X	-	-	Low operating temp.	3.5.1(a)	4.6.2.5
X	X	-	X	High, low operating temp.	3.5.1(a)	4.6.2.14
X	X	-	-	Temperature shock.	3.5.2(b)	4.6.2.6
X	X	-	-	Altitude.	3.5.1(b) and 3.5.2(b)	4.6.2.7
X	X	-	-	Vibration.	3.5.2(f)	4.6.1.6
X	X	-	-	Shock.	3.5.2(e)	4.6.2.9
X	X	-	-	Humidity.	3.5.1(c) and 3.5.2(c)	4.6.2.10
X	X	-	-	Salt fog.	3.5.1(e) and 3.5.2(h)	4.6.2.11

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TABLE I. Test schedule. - Continued

1	2	3	4	5	6	7
X	-	-	-	Fungus growth (Test A).	3.5.2(g)	4.6.2.12.1
-	X	-	-	Fungus growth (Test B).	3.5.2(g)	4.6.2.12.2
X	X	-	-	Electromagnetic interference.	3.6.1.8	4.6.2.13
X	X	-	X	Immersion.	3.5.1(f)	4.6.2.15
X	X	-	X	Final test.	3.6	4.6.2.16
X	X	-	-	Reliability.	3.7	4.6.2.17

4.6.2 Tests.

4.6.2.1 Test conditions. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with Government-approved contractor's test procedures and as specified herein. Calibration of test equipment shall be in accordance with MIL-C-45662. Unless otherwise specified in the individual test procedure, all tests shall be conducted at room temperature (see 6.4.2). Each AAU shall be subjected to vibration on the vertical axis at 0.02 $\pm$ 0.005 inch double amplitude, 60 Hz, for a period of 5 minutes.

4.6.2.2 Burn-In. Each AAU shall be subjected to a 48-hour burn-in. During burn-in the AAU shall be exercised through all of its functions 100 times each 24-hour period. Failures during burn-in shall be analyzed and corrective action taken when appropriate. Copies of failure reports shall be forwarded to the contracting officer. Only AAU units that successfully complete the 48-hour burn-in shall be subjected to the testing specified in 4.6.2.3.

4.6.2.3 Unit test. The following tests shall be performed in order to prove that the electrical characteristics comply with the requirement of this specification. With the AAU connected and provided with appropriate MINISID III detect signals and a power supply that can be adjusted to plus 20 and plus 30 volts, each test shall be conducted at both voltages.

(a) Subject the microphone to the sound pressure level specified in 3.6.1.1 and measure the rms output levels.
(b) With normal work area illumination striking the daylight sensor, the AAU shall be deactivated. Closing the movable light shutter shall permit the normal generation of the audio time control and audio analog signal after appropriate MINISID III detect signals are provided, (see 3.6.1.2).

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(c) With the microphone subjected to low, medium, and high sound pressure levels at the start of each test, examine the audio analog for linearity of amplification and output limit level. Supply sound pressure level for the required system audio frequency response and record levels (see 3.6.1.3).

(d) Connect cable to MINISID III. Plug-in of the AAU shall switch MINISID III from seismic only operation mode to acoustic operation mode (see 3.6.1.4 and 3.6.1.5). Operation of the movable light shutter from daylight to nighttime operation shall activate the AAU.

(e) Provide a series of MINISID detect pulses of 15 ms, plus or minus 5 ms, in length and plus 5.6 to plus 6.3 V in amplitude spaced 10 seconds, plus or minus 2 seconds. Pulse amplitudes and durations specified in 3.6.1.6 shall be observed. The AAU shall respond to the three-pulse signal. Failure to comply with any unit test (a through e) shall constitute failure of this test.

4.6.2.4 High temperature. The AAU, with power on, shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501.1, Procedure I. The highest operating temperature shall be 145° F. Inability of the AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.5 Low temperature. The AAU shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502.1, Procedure I. The storage temperature shall be minus 54° C (minus 65° F), maintained for 8 hours. The operating temperature shall be 0° C (32° F) for 8 hours. Inability of AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.6 Temperature shock. The AAU, with power off, shall be subjected to the temperature shock test in accordance with MIL-STD-810, Method 503.1, Procedure I. Inability of the AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.7 Altitude. The AAU shall be subjected to the altitude test in accordance with MIL-STD-810, Method 500.1, Procedure I, except the altitude shall be 14,000 feet while the AAU is operating and 50,000 feet while the AAU is not operating. Inability to operate at 14,000 feet, inability to operate after exposure to 50,000 feet altitude, any evidence of damage (see 6.4.1), or inability of the AAU to perform as specified in 4.6.2.3 shall constitute failure of this test.

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spring for plate, wire bail

4.6.2.8 Vibration. The AAU, with power off and not contained in its shipping container, shall be subjected to the vibration test with power off in accordance with MIL-STD-810, Method 514.2, Procedure X, Curve AX of Figure 514.2-7. Any evidence of damage (see 6.4.1) or inability of the AAU to perform as specified in 4.6.2.3 shall constitute failure of this test.

4.6.2.9 Shock. The AAU shall be subjected to the following shock tests in accordance with MIL-STD-810.

4.6.2.9.1 Method 516.2, Procedure I, Figure 516.2-1, Amplitude b, time duration d. The AAU shall not be mounted in its shipping container and the power shall be off during this test. Inability of the AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.9.2 Method 516.2, Procedure V. The AAU shall not be in its shipping container and the power shall be on during the test. Inability of the AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.10 Humidity. The AAU, with power OFF, shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I, with the following exception: Sample AAU (see 4.4.2.2) shall be subjected to only 5 cycles of Steps 2, 3, and 4 of Procedure I. Inability of the AAU to perform as specified in 4.6.2.3 or any evidence of damage (see 6.4.1) shall constitute failure of this test.

4.6.2.11 Salt fog. The AAU with all covers, seals, and protective devices removed as required for maintenance settings or adjustments, shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509.1, Procedure I. Any evidence of damage (see 6.4.1) or inability to perform as specified in 4.6.2.3 shall constitute failure of this test.

4.6.2.12 Fungus growth.

4.6.2.12.1 Fungus growth (Test A). The AAU shall be placed within the Tropical Testing Chamber, USAMERADCOM, Fort Belvoir, Virginia (see 6.6), and subjected to exposure under simulated cyclic tropical conditions prevailing therein for a period of ninety (90) days. Environmental conditions within the chamber are as follows:

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The Tropical Testing Chamber is operated on a daily temperature - humidity cycle typical of a tropical area. For an 8-hour period, the temperature is held at 85° F and the relative humidity at 70 percent. During the next eight hours of the cycle, the temperature is gradually lowered to 75° F, and the relative humidity is raised to 95 percent. The temperature and relative humidity are then kept at 75° F and 95 percent for a 4-hour period, followed by a gradual change to 85° F and 70 percent during the next 4-hour period, which completes the cycle. The chamber atmosphere is heavily charged with spores of various species of fungi and provides continuous inoculation of exposed items and materials to such an extent that artificial prior inoculation with fungus spores is unnecessary.

The following list is representative of the fungi present in the chamber:

Acremonium alternatum	Memmoniella echinata
Ascotricha xylina	Mucor sp.
Aspergillus fluvus	Myrothecium verrucaria
Aspergillus niger	Nigrospora sphaerica
Aspergillus oryzae	Penicillium citrinum
Aspergillus terreus	Penicillium luteum
Aspergillus versicolor	Penicillium purpurogenum
Chaetomium bostrychodes	Penicillium stoloniferum
Chaetomium indicum	Pollularia sp.
Chaetomium globosum	Rhizopus nigricans
Coniothyrium sp.	Spicaria sp.
Curvularia sp.	Stachbotrys atra
Fusarium sp.	Trichoderma viride

Upon removal from the chamber after the 90-day exposure period, the AAU shall be examined for damage and tested in accordance with 4.6.2.3. Any evidence of damage or inability to perform as specified in 4.6.2.3 shall constitute failure of this test.

4.6.2.12.2 Fungus Growth (Test B). The AAU shall be tested in accordance with MIL-STD-810, Method 508.1, Procedure I. After conclusion of the test, the AAU shall be examined for damage (see 6.4.1). The AAU shall then be subjected to all tests specified in Column 3 of Table I. Inability to pass any of these tests or evidence of damage (see 6.4.1) shall constitute failure of this test.

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4.6.2.13 Electromagnetic interference. The preproduction model, equipped for the reduction of electromagnetic interference in accordance with 3.6.1.8, shall be tested in accordance with MIL-STD-461. The contractor shall furnish the contracting officer the test report required by MIL-STD-461 within the time frame specified (see 6.2). Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production units. Disapproval of the report shall constitute failure of this test.

4.6.2.14 High/low operating temperature. The AAU, with power on only at operating temperature ranges, shall be subjected to the following high and low temperature cycle shock tests. Any evidence of damage (see 6.4.1) or inability to perform as specified in 4.6.2.3 shall constitute failure of this test.

(a) Place the AAU in the test chamber and activate. Raise the temperature to plus 145° F and maintain for four hours. Perform an abbreviated operational test and then remove the batteries.
(b) Raise the temperature to plus 160° F and maintain for four hours.
(c) Reduce the temperature at a rate of not less than 45° F per minute to minus 65° F and maintain for four hours.
(d) Raise the temperature at a rate of not less than 45° F per minute to plus 160° F and maintain for four hours.
(e) Reduce the temperature to plus 32° F and maintain for one (1) hour. At this point, reinstall the batteries and activate the AAU. Resume test and maintain the temperature at plus 32° F for three hours. Perform an abbreviated operational test.
(f) Raise the temperature to ambient room temperature (see 6.4.2) and maintain for one hour. At the conclusion of this test, the AAU shall be subjected to a complete operational test.

4.6.2.15 Immersion. The AAU shall be subjected to the leakage test in accordance with MIL-STD-810, Method 512, Procedure I, except that the temperature difference between the water and the AAU at the time of submersion shall be not less than 10° C. The water shall be cooler. Evidence of water leakage, damage (see 6.4.1), or inability to perform as specified in 4.6.2.3 shall constitute failure of this test.

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4.6.2.16 Final test. Each AAU shall be tested with a complete MINISID III and a receiver/decoder equivalent to USQ-46. This shall be a Go/No Go type of test. The transmission time shall be in accordance with the requirements of this specification and the audio shall be intelligible. Failure of this test shall be cause for rejection and shall be cause for refusal by the Government to continue acceptance of production AAU's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies.

4.6.2.17 Reliability test. When specified (see 3.7), the AAU shall be tested in accordance with MIL-STD-781, Test Plan IVA using the MTBF specified in 3.7. During this period of operation, the AAU shall be subjected to 100 activations per day. When this reliability test is to be conducted, the time frame for conducting this test, the number of AAU units to be tested, and who shall perform the tests shall be as specified (see 6.2). The test samples shall not be exposed to the shock, salt fog, and fungus growth prior to the reliability testing. Items selected for reliability testing shall have been burned-in as specified in 3.6.1.9 and shall have passed the quality conformance inspection specified in 4.5. The performance of the AAU battery shall not be tested. Periodic performance checks (see 4.6.2.3) shall be made throughout the reliability test. The tests shall be made under ambient environmental laboratory conditions. Nonconformance to 3.7 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select AAU's at any time during the contract production period and subject these AAU's to the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production AAU's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison AAU to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison AAU and shall be cause for refusal by the Government to continue acceptance of production AAU's until evidence has been provided by the contractor that corrective action has been taken to

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eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on AAU's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all AAU's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following possible defects. AQL shall be 1.0 percent defective for major defects and 2.5 percent for minor defects.

Major

110. Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
111. Contents not cushioned within the box as specified for Level A.
112. Box not closed and sealed by method specified for Level A.
113. Quantities packed together exceed the size or weight limitation specified for Level A or B.
114. Strapping not as specified for Level A.

Minor

204. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

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5.1.1 Level A. The AAU assembly and microphone contained in the carrying case shall be preserved in accordance with MIL-P-116, Method IIb. The inner container shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125. The contents shall be cushioned and secured within the box in a manner to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c. Closure and sealing of each of the boxes shall be in accordance with the appendix to the box specification, using Method V on the exterior box and Method I or II on the inner box.

5.1.2 Commercial. The AAU assembly and microphone contained in the canvas carrying case shall be preserved in accordance with FED. STD. No. 356.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The AAU assemblies, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, Grade B, in quantities not to exceed 200 pounds. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The AAU assemblies, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The AAU assemblies, preserved as specified in 5.1, shall be packed in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with FED. STD. No. 356.

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6. NOTES

6.1 Intended use. The AAU is a hand-emplaced recoverable ancillary unit to the MINISID III and is intended for use in detecting acoustic signals resulting from movement of vehicles and personnel. The primary function of the AAU is to detect audible disturbances in the vicinity of MINISID III and to supply verification of intrusion criteria detected by the seismic detector in MINISID III. In a second mode of operation, the AAU serves only as an auxiliary battery for extending the function life of MINISID III.

6.1.1 Implant parameter. The AAU employs an external microphone which may be affixed to some object up to 10 feet from the AAU. A daylight sensor and movable cover mounted in the microphone housing will serve to disable the AAU only during daylight hours if desired. When the AAU is to serve only as an auxiliary battery for MINISID III, the microphone is physically disconnected. When the AAU is disconnected from the MINISID III, all internal devices and circuits are completely deactivated and current drain will be zero. The methods of function switching described above eliminate the need for external switches.

6.1.2 Field settings. There are no external switches or controls except the light sensor cover. Function switching is automatic upon disconnection of the microphone cable from the connector on the AAU, or upon disconnection of the cable to MINISID III.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of AAU's to be furnished (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examintion and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and number of AAU's to be furnished, when applicable (see 3.3).
(e) When reliability test is required, the time frame for conducting the reliability test, the number of AAU units to be tested, and who shall perform the tests (see 3.7 and 4.6.2.17).

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(f) Time frame required for submission of electromagnetic interference test report (see 4.6.2.13).
(g) Degree of preservation and packing required (see 5.1 and 5.2).
(h) When strapping shall be other than as specified (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production AAU's from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligations to furnish AAU's conforming to this specification.

6.4 Definitions.

6.4.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance with the AAU with the provision of this specification.

6.4.2 Room temperature. Room temperature is 75° F, plus or minus 5° F.

6.4.3 Temperature shock. Changes in temperature up to 50° F per minute.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.8.

6.6 Tropical testing chamber. Arrangements should be made to make the Tropical Testing Chamber, USAMERADCOM, Fort Belvoir, Virginia, available to the contractor for conducting the fungus growth test (see 4.6.2.12).

Custodians:
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Navy - SH

Preparing activity:
Army - ME

Project 6350-0039

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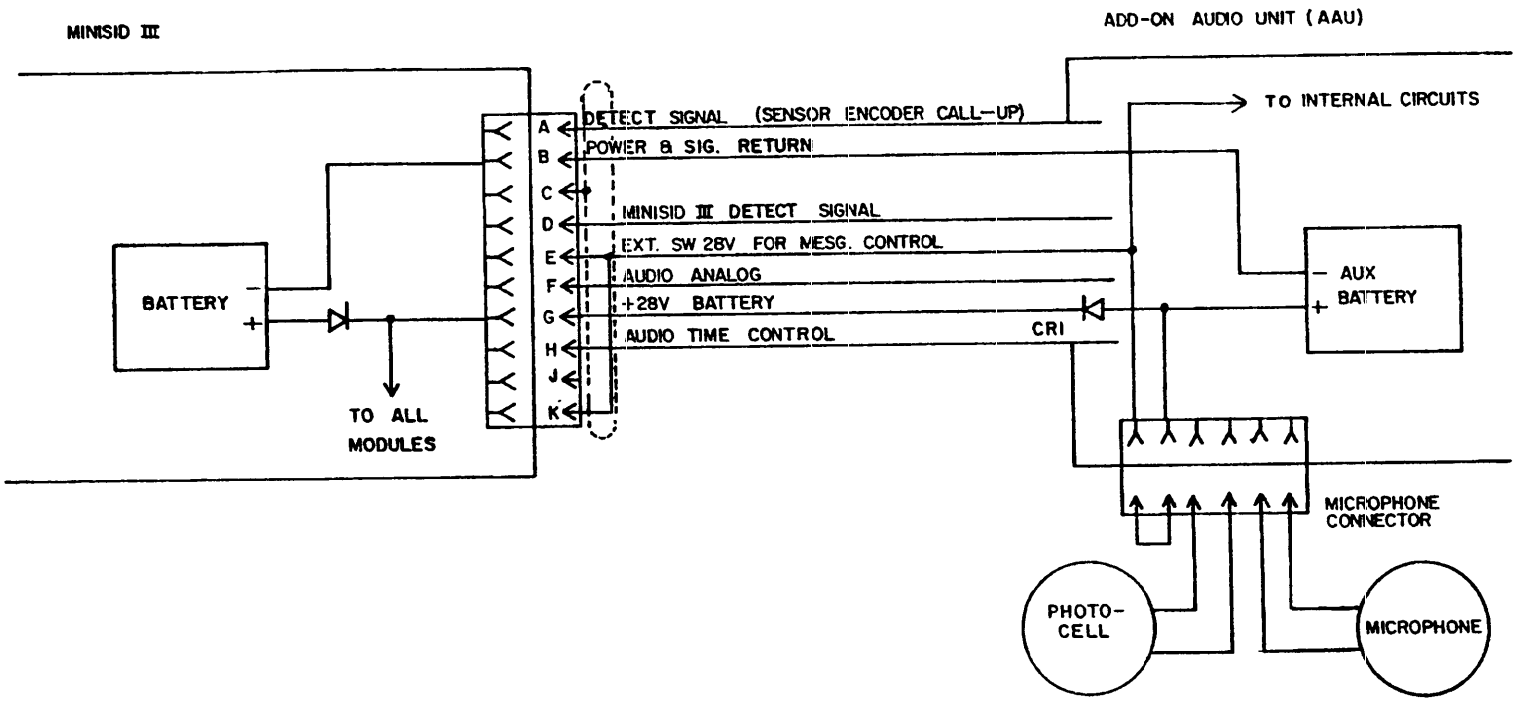


FIGURE I. AAU/MINISID III INTERFACE

X-2639A

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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